

## SMIDOVICH, Russian Federation

WMO: 317250

Lat: 48.617N

Lon: 133.833E

Elev: 172

StdP: 14.61

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -26.7      | -23.0 | -31.2                         | 1.0 | -25.5 | -27.5 | 1.2 | -21.9 | 18.2                  | 2.9  | 14.3 | 4.7  | 2.2                      | 230  | 0.540 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.6                   | 85.6            | 73.3 | 82.7 | 71.2 | 80.1 | 69.6 | 75.1                | 82.4 | 73.3 | 80.2 | 71.5 | 77.7 | 5.8                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.8                            | 122.4 | 79.4 | 71.0 | 114.8 | 77.6 | 69.3 | 108.2 | 75.6 | 38.8          | 82.1 | 37.2 | 80.3 | 35.5 | 77.9 | 85.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |      | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 17.6 | 14.0 | 11.8                       | DB  | -31.3              | 90.7 | 4.9                                                | 2.9 | -34.8      | 92.9 | -37.6      | 94.6 | -40.3      | 96.2 | -43.9 | 98.4 |
| (5)               |      |      |                            | WB  | -31.4              | 78.3 | 4.8                                                | 2.6 | -34.9      | 80.2 | -37.7      | 81.7 | -40.4      | 83.2 | -43.9 | 85.1 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 35.4  | -6.0  | 1.5   | 18.7  | 39.9  | 54.6  | 64.5  | 70.6  | 67.6  | 56.4  | 40.1  | 17.5  | -2.7 | (6)  |
| (7)  |                                                                   | DBStd       | 28.39 | 9.20  | 8.65  | 10.62 | 7.58  | 6.93  | 5.87  | 4.57  | 4.98  | 6.54  | 7.64  | 11.32 | 9.52 | (7)  |
| (8)  |                                                                   | HDD50       | 7358  | 1736  | 1358  | 969   | 316   | 34    | 0     | 0     | 0     | 19    | 318   | 974   | 1634 | (8)  |
| (9)  |                                                                   | HDD65       | 11174 | 2201  | 1778  | 1434  | 754   | 330   | 78    | 8     | 30    | 265   | 773   | 1424  | 2099 | (9)  |
| (10) |                                                                   | CDD50       | 2029  | 0     | 0     | 0     | 12    | 176   | 436   | 640   | 546   | 209   | 10    | 0     | 0    | (10) |
| (11) |                                                                   | CDD65       | 370   | 0     | 0     | 0     | 0     | 6     | 64    | 183   | 111   | 6     | 0     | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 2728  | 0     | 0     | 0     | 7     | 108   | 581   | 1305  | 675   | 51    | 1     | 0     | 0    | (12) |
| (13) |                                                                   | CDH80       | 616   | 0     | 0     | 0     | 1     | 12    | 143   | 350   | 107   | 3     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.9   | 4.1   | 4.4   | 5.3   | 6.1   | 5.9   | 4.9   | 4.3   | 4.3   | 4.6   | 5.4   | 5.4   | 4.4  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 26.3  | 0.4   | 0.4   | 0.9   | 1.5   | 2.6   | 3.1   | 5.7   | 5.1   | 3.3   | 1.7   | 0.9   | 0.6  | (15) |
| (16) |                                                                   | PrecMax     | 36.8  | 1.2   | 1.6   | 2.3   | 4.2   | 5.5   | 8.1   | 14.1  | 9.7   | 7.6   | 4.4   | 3.0   | 2.7  | (16) |
| (17) |                                                                   | PrecMin     | 18.2  | 0.0   | 0.0   | 0.0   | 0.1   | 0.5   | 0.4   | 0.9   | 1.3   | 0.6   | 0.3   | 0.0   | 0.1  | (17) |
| (18) |                                                                   | PrecStd     | 4.4   | 0.3   | 0.4   | 0.7   | 0.9   | 1.3   | 2.0   | 2.9   | 2.0   | 1.7   | 1.1   | 0.7   | 0.6  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 23.2  | 32.2  | 49.8  | 72.6  | 81.6  | 88.7  | 90.6  | 86.1  | 78.8  | 69.2  | 46.9  | 28.8 | (19) |
| (20) |                                                                   | MCWB        | 19.8  | 27.1  | 38.8  | 54.6  | 63.3  | 72.2  | 74.8  | 75.6  | 67.6  | 54.7  | 39.8  | 25.9  | (20) |      |
| (21) |                                                                   | 2%          | DB    | 17.8  | 25.1  | 42.6  | 64.2  | 77.2  | 83.7  | 87.0  | 83.0  | 74.9  | 61.9  | 41.1  | 20.2 | (21) |
| (22) |                                                                   | MCWB        | 15.7  | 21.5  | 34.9  | 50.7  | 59.5  | 70.1  | 74.1  | 72.6  | 64.5  | 50.3  | 35.9  | 18.2  | (22) |      |
| (23) |                                                                   | 5%          | DB    | 12.4  | 20.4  | 38.1  | 58.3  | 72.8  | 80.4  | 83.8  | 80.4  | 71.8  | 57.1  | 36.6  | 15.2 | (23) |
| (24) |                                                                   | MCWB        | 10.6  | 17.7  | 32.0  | 46.1  | 57.6  | 67.6  | 72.8  | 70.7  | 62.5  | 47.2  | 32.5  | 13.5  | (24) |      |
| (25) |                                                                   | 10%         | DB    | 8.1   | 16.0  | 34.5  | 53.1  | 68.8  | 76.8  | 80.8  | 77.8  | 68.6  | 52.8  | 33.1  | 11.5 | (25) |
| (26) |                                                                   | MCWB        | 6.7   | 13.9  | 29.6  | 43.0  | 55.8  | 66.0  | 71.2  | 69.1  | 60.7  | 44.7  | 30.1  | 10.0  | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 20.7  | 26.5  | 39.7  | 55.9  | 66.1  | 75.3  | 77.5  | 77.4  | 69.6  | 56.5  | 40.6  | 26.3 | (27) |
| (28) |                                                                   | MDCB        | 22.7  | 32.0  | 48.3  | 69.3  | 79.3  | 85.0  | 86.3  | 84.2  | 75.9  | 65.2  | 45.1  | 29.2  | (28) |      |
| (29) |                                                                   | 2%          | WB    | 15.6  | 21.8  | 35.7  | 51.1  | 61.9  | 71.6  | 75.5  | 74.8  | 66.8  | 52.2  | 36.3  | 18.5 | (29) |
| (30) |                                                                   | MDCB        | 17.7  | 25.1  | 41.6  | 62.3  | 73.1  | 81.2  | 83.8  | 80.3  | 72.8  | 59.3  | 40.3  | 20.3  | (30) |      |
| (31) |                                                                   | 5%          | WB    | 10.6  | 17.8  | 32.5  | 47.5  | 59.2  | 69.2  | 73.9  | 72.8  | 64.2  | 48.8  | 33.4  | 13.8 | (31) |
| (32) |                                                                   | MDCB        | 12.1  | 20.4  | 37.2  | 56.8  | 69.7  | 78.1  | 81.5  | 78.1  | 69.5  | 54.9  | 36.2  | 15.2  | (32) |      |
| (33) |                                                                   | 10%         | WB    | 6.6   | 14.0  | 29.9  | 44.0  | 56.9  | 67.2  | 72.4  | 70.8  | 61.9  | 45.6  | 30.3  | 10.1 | (33) |
| (34) |                                                                   | MDCB        | 8.0   | 16.0  | 34.3  | 51.8  | 66.0  | 75.0  | 79.1  | 75.7  | 67.1  | 51.6  | 33.4  | 11.5  | (34) |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 18.7  | 21.2  | 19.8  | 18.9  | 19.9  | 18.5  | 16.6  | 16.6  | 20.0  | 20.1  | 16.1  | 16.6  | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 20.0  | 22.6  | 20.4  | 28.5  | 28.0  | 25.1  | 21.9  | 20.4  | 24.7  | 27.5  | 18.3  | 16.9 | (36) |
| (37) |                                                                   | MCWBR       | 18.2  | 19.3  | 13.9  | 16.3  | 13.5  | 12.6  | 10.8  | 11.0  | 14.1  | 16.1  | 13.4  | 15.3  | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 19.9  | 22.3  | 19.5  | 26.5  | 23.8  | 21.8  | 19.0  | 17.2  | 20.0  | 24.6  | 17.6  | 16.5 | (38) |
| (39) |                                                                   | MCWBR       | 18.0  | 19.1  | 13.7  | 15.8  | 12.8  | 11.8  | 10.4  | 10.3  | 12.7  | 15.8  | 13.3  | 15.0  | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.284 | 0.326 | 0.401 | 0.543 | 0.567 | 0.520 | 0.507 | 0.448 | 0.413 | 0.438 | 0.362 | 0.297 | (40) |      |
| (41) |                                                                   | taud        | 2.157 | 2.010 | 1.876 | 1.729 | 1.761 | 1.968 | 2.050 | 2.216 | 2.218 | 1.976 | 2.073 | 2.126 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 240   | 252   | 249   | 225   | 226   | 240   | 241   | 251   | 247   | 207   | 201   | 214   | (42) |      |
| (43) |                                                                   | Edh at Noon | 26    | 38    | 52    | 68    | 68    | 56    | 51    | 41    | 37    | 40    | 29    | 23    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 592   | 989   | 1442  | 1529  | 1672  | 1777  | 1640  | 1486  | 1226  | 834   | 550   | 458   | (44) |      |
| (45) |                                                                   | RadStd      | 19    | 30    | 102   | 158   | 147   | 238   | 155   | 105   | 110   | 63    | 39    | 26    | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page